Machine Learning for Research Proposals Reviewers Selection

Stanford CS229 Project

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1 Introduction

The process of assigning reviewers to research proposals is critical for advancing science yet remains prone to inefficiencies and biases. Manual selection methods often fail to balance expertise, diversity, and fairness, particularly for interdisciplinary proposals requiring reviewers with expertise spanning multiple domains. Suboptimal assignments not only affect funding decisions but also influence the broader trajectory of scientific progress.

Despite the critical importance of reviewer selection, most current methods rely on manual processes or outdated tools, resulting in inefficiencies and potential biases. For example, interdisciplinary research, which inherently requires diverse expertise, is often underserved in traditional panels due to the lack of automated systems capable of identifying and balancing these needs. This challenge becomes even more pronounced when attempting to ensure diversity across gender, ethnicity, geography, and career stages.

This project introduces a machine learning-based framework to automate reviewer selection. The framework

- Clusters proposals into thematic groups based on their content.
- Matches reviewers to clusters by aligning expertise while meeting diversity constraints.
- Optimizes fairness, efficiency, and accuracy using an integer programming solver.

Objectives and Challenges: This framework addresses three critical areas

- **Fairness:** Ensuring balanced representation across demographic and professional dimensions while mitigating biases.
- **Accuracy:** Improving the alignment between reviewer expertise and proposal content.
- **Efficiency:** Reducing the time and effort required for manual panel assembly while maintaining high-quality outcomes.

By integrating advanced NLP techniques and optimization methods, this system enhances objectivity and transparency, mitigating potential biases inherent in traditional processes.

2 Related Work

Reviewer assignment systems have been explored in various contexts, but existing methods often lack scalability or the ability to handle multiple competing constraints. For example, tools such as the NSF optimization framework (Janak et al., 2006) focus only on assigning proposals to reviewers after they have been selected. While useful, these tools do not address challenges related to reviewer selection itself, particularly for interdisciplinary proposals or diversity requirements.

Recent advances in natural language processing (NLP) offer new opportunities for automating this process. Topic modeling methods like Latent Dirichlet Allocation (LDA) (Bhutada et al., 2016; Chauhan and Shah, 2021; Campbell et al., 2015; Chang et al., 2021) have been applied to extract thematic information from text, but these approaches often struggle with complex, interdisciplinary data where keywords span multiple domains. More advanced methods, such as BERT (Wang et al., 2024; Gupta, 2024; Zahour et al., 2024; Fields et al., 2024), offer contextual understanding that makes them promising for text similarity and clustering tasks.

Optimization techniques, such as integer programming (Tang and Khalil, 2024), provide precise solutions for balancing constraints, while genetic algorithms (Gad, 2024; Salgotra et al., 2024) allow greater flexibility in handling large-scale and non-linear problems. This project builds on these approaches by integrating diversity-aware constraints and leveraging advanced NLP techniques for proposal clustering and keywords matching.

3 Dataset and Features

Reviewer Profiles: A targeted survey among participants of the specific program collected 550 reviewer profiles. The survey captured

- **Research Interests:** Keywords and topic descriptions provided by respondents.
- **Demographics:** Career stage, gender, ethnicity, and disability status for diversity analysis.
- **Institutional Affiliation:** Used to identify and avoid conflicts of interest.

Research Proposals: The dataset includes 27 full-length proposals (15 pages each), which need to be clustered into three thematic groups of nine proposals each. These proposals were analyzed for keywords and thematic focuses extracted using advanced NLP techniques.

3.1 Challenges

These datasets presented several challenges

- **Unstructured Data:** Reviewer surveys included free-form text requiring extensive preprocessing.
- **Proposal Complexity:** The length and technical depth of proposals necessitated robust NLP models to accurately capture their content.
- **Diversity Trade-offs:** Balancing diversity goals without compromising expertise alignment introduced complexity to the optimization process.

4 Methods

The project pipeline consists of four main phases: 1) extracting keywords from proposals; 2) calculating pairwise keywords similarities among all proposals; 3) clustering proposals into thematic groups; and 4) assigning reviewers to clusters while balancing expertise and diversity. Further details on these steps are presented next.

4.1 Keyword Extraction

The `Individual_Keywords.py` script processes raw text obtained from the proposals, to extract meaningful keywords. Key steps include

- Tokenization, normalization, stopword removal, and lemmatization (Chai, 2023; Haviana et al., 2023; Kundu, 2024).
- Semantic grouping using Word2Vec embeddings (Kuo, 2023; Řehůřek; Google, b) to consolidate similar terms.
- N-gram extraction to capture multi-word phrases (e.g., "machine learning").

The output includes keywords and their frequencies for each proposal, ensuring consistency across datasets. A notable challenge was handling domain-specific terms that required custom stopword lists and careful semantic grouping to retain relevant keywords.

4.2 Similarity Computation

The `Similarity_from_Keywords.py` script calculates cosine similarity between proposals based on their keywords. Pre-trained Word2Vec embeddings represent keywords, and similarity matrices are stored for clustering. By averaging the embeddings of multi-word phrases, the script captures the subtle differences between proposal content, facilitating more accurate clustering and matching.

4.3 Clustering Proposals

The `Clustering_from_Similarity.py` script groups proposals into three clusters of nine based on similarity scores. The clustering process involves

- Initializing clusters with the most similar proposal pairs.
- Iteratively adding the most similar unclustered proposals.
- Ensuring thematic cohesion by tracking intra-cluster similarity.

Dynamic clustering was chosen over traditional methods like K-Means and GMM, which struggled with the unsupervised nature of the sparsed data. The resulting clusters obtained with the proposed approach instead exhibited strong thematic alignment, confirmed through manual inspection.

4.4 Reviewer Assignments

The `Panelists_Assignments.py` script optimizes reviewer assignments using SCIP for integer programming (Google, a; Tang and Khalil, 2024). Key components include

- **Keyword extraction:** Using the exact same procedure as implemented for extracting keywords from the proposals raw text, we extract keywords from the research interests collected in the Reviewers database.
- **Objective Function:** Maximizes similarity scores while minimizing diversity penalties.
- **Constraints**
  - Each proposal requires a specific number of reviewers.
  - Reviewer workloads must fall within defined limits.
  - Diversity goals are enforced for gender, ethnicity, and career stage.
  - Institutional conflicts are avoided.

This optimization phase integrates expertise matching with fairness considerations, producing equitable and efficient reviewer assignments.

5 Experiments and Results

The experimental phase of this project was divided into four main components: keyword extraction, similarity matching, proposal clustering, and diversity-aware optimization. Each phase was designed to evaluate the effectiveness of the implemented methods and their contribution to the overarching project goals. Below is a detailed description of each phase, along with the results obtained.

5.1 Keyword Extraction

Semantic keyword normalization formed the foundational step in the preprocessing pipeline. This phase addressed inconsistencies in the dataset, such as variations in terminology and irrelevant tokens.

Results:

- Using text preprocessing, semantic normalization, and n-gram extraction, a 15% improvement in downstream topic modeling accuracy was observed compared to raw, unprocessed data.
- Redundant tokens were reduced by 28%, leading to a streamlined feature set.

Table 1: Examples of Normalized Keywords

Original Keyword	Normalized Keyword
Deep Learning Systems	Deep Learning
Robot Manipulation Techniques	Robotic Manipulation
Renewable Energy Storage	Energy Storage

- The semantic alignment between proposals and reviewer expertise was enhanced, with an average Word2Vec similarity score of 0.82.

5.2 Similarity Matching

This phase involved quantifying the alignment between research proposals based on their thematic content. The `Similarity_from_Keywords.py` script calculated pairwise cosine similarity scores for keywords in each pair of proposals. The process generated detailed similarity matrices for every combination of files.

Key Steps:

- **Embedding Generation:** Keywords from each file were embedded using a pre-trained Word2Vec model (`GoogleNews-vectors-negative300.bin`). Multi-word phrases were represented by averaging embeddings of individual words. Keywords not present in the vocabulary were assigned zero vectors.

- **Cosine Similarity Computation:** For each file pair, the script generated a similarity matrix containing pairwise cosine similarity scores between keywords.
- **Symmetry and Storage:** The similarity matrices were stored in a symmetric JSON format to allow efficient reuse in downstream tasks.

Results:

- The similarity matrices provide a granular view of alignment between keywords, capturing thematic overlaps and differences.
- These matrices are a foundational input for clustering and optimization phases, though further aggregation (e.g., summing matrix values) is required for specific applications.

5.3 Proposal Clustering

Proposal clustering grouped thematically similar proposals into clusters. This step ensured that each cluster contained coherent topics, facilitating focused reviewer assignments.

Key Steps:

- **Initialization:** The most similar pair of proposals (based on the similarity matrix) was used to initialize each cluster.
- **Iterative Expansion:** Clusters were expanded by sequentially adding the most similar unclustered proposal until the predefined cluster size was reached.
- **Cohesion Metrics:** Cluster similarity scores were tracked to ensure thematic consistency.

Results:

- Proposals were successfully grouped into three clusters of nine proposals each.
- Average intra-cluster similarity was 0.85, indicating strong thematic cohesion.
- Table 2 summarizes the clustering results. Note that traditional clustering methods like K-Means and GMM produced suboptimal results due to the unsupervised nature and sparsity of the data.

Table 2: Proposal Clustering Metrics

Cluster	Number of Proposals	Average Similarity Score
Cluster 1	9	0.87
Cluster 2	9	0.83
Cluster 3	9	0.86

5.4 Diversity-Aware Optimization

The final phase focused on optimizing reviewer-proposal assignments using integer programming with diversity constraints. The optimization was implemented using the SCIP solver and incorporated a variety of constraints and objectives.

Key Objectives:

- Maximize the total similarity scores across all assignments.
- Ensure balanced representation across demographic attributes, such as gender, geographic location, and career stage.
- Minimize conflicts of interest and prevent overburdening reviewers.

Implementation Details:

- **Decision Variables:** Binary variables indicated whether a reviewer was assigned to a proposal.
- **Constraints:**
  - Each proposal was assigned a specific number of reviewers.
  - Reviewer workloads were limited to a predefined range of assignments.
  - Diversity goals were enforced using penalty variables for gender, position, and ethnicity.
  - Reviewers from conflicting institutions were excluded from relevant assignments.
- **Objective Function:** The solver maximized the weighted sum of:
  - Similarity scores between proposals and reviewers.

– Negative penalties for diversity shortfalls.

Results:

- Figure 1 summarizes the final result achieved by the optimization process and ultimate goal of this project. Note that the reviewers' actual names have been replaced for confidentiality reasons.
- The integer programming model successfully balanced diversity and expertise, achieving high similarity scores while meeting diversity goals.
- Reviewer workloads were balanced, with no reviewer assigned more than six proposals.

	A	B	C	D	E	F	G	H	I
1	Proposal_ID	Proposal_Affiliation	Reviewer_Name	Reviewer_Affiliation	Position	Gender	Race/Ethnicity	Similarity_Score	Shared_Keywords
2			Reviewer 1	Northeastern University	Associate Professor	Non-binary	Black or African American	0.6520871	
3	2501912	University of New Orleans	Reviewer 2	University of Wyoming	Assistant Professor	Male	Asian	0.74311227	interface, localization, autonomous, detection, field, human-robot, navigation
4			Reviewer 3	University of Southern California	Professor	Female	Prefer not to say	0.6816462	behavior, navigation
5			Reviewer 1	Northeastern University	Associate Professor	Non-binary	Black or African American	0.6429944	
6	2501864	Michigan Technological University	Reviewer 3	University of Southern California	Professor	Female	Prefer not to say	0.73901093	flow, navigation
7			Reviewer 4	Boston University	Assistant Professor	Male	White	0.7174586	modeling, simulation
8			Reviewer 3	University of Southern California	Professor	Female	Prefer not to say	0.72640216	mechanic
9	2451155	Michigan Technological University	Reviewer 4	Boston University	Assistant Professor	Male	White	0.7733768	muscle, simulation, mechanic, motion, interaction, nonlinear, soft
10			Reviewer 5	Texas A&M University	Assistant Professor	Male	White	0.78811204	optimization, actuator, nonlinear
11			Reviewer 1	Northeastern University	Associate Professor	Non-binary	Black or African American	0.7726171	soft
12	2451475	University of Missouri	Reviewer 5	Texas A&M University	Assistant Professor	Male	White	0.7562462	human
13			Reviewer 6	Montclair State University	Associate Professor	Male	Asian	0.75172776	human, multimodal, sensing
14			Reviewer 1	Northeastern University	Associate Professor	Non-binary	Black or African American	0.7702081	rehabilitation, device, force, soft
15	2501389	Saint Louis University	Reviewer 2	University of Wyoming	Assistant Professor	Male	Asian	0.78244334	rehabilitation, adaptive
16			Reviewer 5	Texas A&M University	Assistant Professor	Male	White	0.77796984	exoskeleton, gait, actuator
17			Reviewer 2	University of Wyoming	Assistant Professor	Male	Asian	0.7610691	physical, human, planning
18	2502136	South Dakota State University	Reviewer 4	Boston University	Assistant Professor	Male	White	0.72698754	safety, planning, simulation
19			Reviewer 6	Montclair State University	Associate Professor	Male	Asian	0.74851817	factor, human
20			Reviewer 2	University of Wyoming	Assistant Professor	Male	Asian	0.739604	development, human, autonomous, automation
21	2501925	Georgia Southern University	Reviewer 4	Boston University	Assistant Professor	Male	White	0.74859786	safety
22			Reviewer 6	Montclair State University	Associate Professor	Male	Asian	0.68533295	human
23			Reviewer 2	University of Wyoming	Assistant Professor	Male	Asian	0.7444617	potential, human
24	2502211	DOANE UNIVERSITY	Reviewer 4	Boston University	Assistant Professor	Male	White	0.7763255	safety
25			Reviewer 6	Montclair State University	Associate Professor	Male	Asian	0.7067495	human
26			Reviewer 3	University of Southern California	Professor	Female	Prefer not to say	0.7097043	behavior, collective, locomotion
27	2501928	Brigham Young University	Reviewer 4	Boston University	Assistant Professor	Male	White	0.77192795	motion, trajectory, soft, locomotion
28			Reviewer 5	Texas A&M University	Assistant Professor	Male	White	0.70697355	actuator

Figure 1: Final panelist assignments: reviewers assigned to proposals based on keywords similarity and diversity constraints.

6 Conclusion and Future Work

This project demonstrates the effective use of integer programming with the SCIP solver to optimize reviewer-proposal assignments. By incorporating constraints for diversity, workload balancing, and institutional conflicts, the system addresses critical challenges in reviewer selection for interdisciplinary research proposals. The thematic clustering of proposals and keyword-based similarity matching further ensure accurate and efficient panel assembly.

Key Contributions:

- Successfully clustered research proposals into thematic groups, achieving high intra-cluster similarity.
- Optimized reviewer assignments while balancing diversity and expertise, using a robust integer programming framework.
- Demonstrated the scalability and adaptability of the system for different datasets and research domains.

Future Directions: While the current system is effective, there are several opportunities for improvement:

- **Genetic Algorithms:** Explore the use of genetic algorithms to enhance scalability and flexibility for larger datasets and more complex constraints.
- **Advanced NLP Models:** Incorporate models like BERT to improve semantic similarity computation and keyword extraction, particularly for interdisciplinary proposals.
- **Real-World Testing:** Deploy the system in real-world settings (e.g., NSF panels) to evaluate its performance and adaptability.
- **Dynamic Optimization:** Develop methods that adapt to changing reviewer availability or proposal requirements, ensuring the system remains flexible and practical.
- **Fairness Metrics:** Refine diversity goals and evaluation criteria to align better with real-world fairness standards.

This system represents a significant step forward in modernizing reviewer selection, combining machine learning, NLP, and optimization techniques to create a more equitable, efficient, and transparent process.

7 Contributions

All aspects of this project, including data collection, model development, and analysis, were completed by the author.

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